

Tejal Anavekar

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

Aug 2024 – May 2026

- Relevant skills: Analysis of Algorithm, Database management, Applied Natural Language Processing, Machine Learning for Data Science, Information Retrieval, Geospatial Information Management

Fr.C.Rodrigues Institute of Technology

Mumbai, India

Bachelor of Engineering in Information Technology

July 2020 – June 2024

- Relevant skills: Operating Systems, Computer Networks, Machine Learning, Artificial Intelligence, Data Science, Data Mining, Big Data

TECHNICAL SKILLS

Programming: Python, C, C#, HTML, CSS, Javascript

Systems & OS: Linux/Unix, Shell Scripting, TCP/IP, HTTP/HTTPS

Framework: ASP.NET Core, .NET, ReactJs, NodeJs, ExpressJS, APIs

Libraries: PyTorch, Scikit-learn, Keras, OpenCV, TensorFlow, Bootstrap, Pandas, NumPy, Seaborn, Matplotlib

Databases & Cloud : SQL, Mongo DB, PostgreSQL, Firebase, AWS (EC2, S3), GCP

EXPERIENCE

Software Developer

Sept 2024 – Present

USC Institute for Creative Technologies

Los Angeles, CA

- Developed ICT internal web pages using C# Razor Pages, replicating reference UI designs, and used C# Entity Framework Core to generate Excel reports from production data deployed to live environments.
- Debugged and resolved cross-service frontend-backend failures using JavaScript, improving production system reliability.
- Built and supported production monitoring dashboards and backend data services, integrating **real-time data streams** to track critical financial KPIs

Data Engineer Intern

June 2025 – Aug 2025

Meta Platforms, Inc

Burlingame, CA

- Designed and implemented **4** scalable pipelines for distributed data processing and handling **billions of records** to power real-time analytics and AI readiness dashboards in production environments
- Built automated validation and processing frameworks in Python to ensure scalability, fault tolerance, and reliable production data workflows
- Refactored and optimized computation logic to reduce processing overhead and improve performance, leading to adoption across **3+** cross-functional engineering teams

AI Engineer

June 2022 – Apr 2023

Tata Institute of Fundamental Research

Mumbai, India

- Applied advanced image segmentation and **SIFT-based feature extraction**, boosting classification precision by **25%** and enabling more precise data interpretation
- Trained and deployed **Convolutional Neural Networks** (ResNet, VGG variants) to automate image classification tasks, improving accuracy and cutting processing time by 30%
- Designed modular **2** preprocessing pipelines to standardize image resolution and brightness for robust ML training across diverse datasets and imaging conditions

PROJECTS | PUBLICATIONS

Budget Buddy | MERN, LangChain

Nov 2025 – Present

- Built a full-stack MERN expense management system with RESTful APIs, MongoDB schema, and AI-powered financial insights, developed an interactive dashboard with real-time monthly spending charts and category-based segmentation, improving users' budget tracking efficiency by 35%

Query Bot | NLP, BERT, LSTM, Pytorch

Jan 2025 – Apr 2025

- Led the development of a custom seq2seq model combining **BERT encoders and LSTM decoders** with schema-aware attention for translating natural language questions to SQL queries. Achieved 91.8% execution accuracy and 82.2% exact match on Spider dataset

GeoForecast - Weather App | MERN Stack, Flask, Swift, GCP

Aug 2024 – Dec 2024

- Designed and deployed a **modular microservices-based backend architecture** using Node.js and MongoDB, hosted on GCP with scalable API integrations. Integrated Google and Tomorrow.io APIs for geolocation, weather prediction and daily forecasts, achieving 93% location accuracy and increasing daily retention by 35%

Smart Surveillance System for Intrusion Detection | SSD, YOLO, Open CV, Tensorflow

July 2023 – Apr 2024

- Devised and Implemented AI-powered smart surveillance system utilizing **SSD and YOLO** algorithms for real-time leopard detection across 500 meters, achieving 92% detection accuracy with false positive reduction of 30%
<https://rjpn.org/jetnr/papers/JETNR2404011.pdf>